



Recitation 6

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1. Homework

2. Review

3. Examples

4. Recitation



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2. Computational problem. Here is a probabilistic method for computing the area of a given subset S of the unit square. The method uses a sequence of independent random selections of points in the unit square $[0, 1] \times [0, 1]$, according to a uniform probability law. If the i th point belongs to the subset S the value of a random variable X_i is set to 1, and otherwise it is set to 0. Let $X_1, X_2, \dots$ be the sequence of random variables thus defined, and for any n , let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

(c) Write a computer program to generate S_n for $n = 1, 2, \dots, 10000$, using the computer's random number generator, for the case where the subset S is the circle inscribed within the unit square. How can you use your program to measure experimentally the value of π ?

Note 1: We can use the formula derived in (b) to iterate.

$$S_n = \frac{n-1}{n} S_{n-1} + \frac{1}{n} X_n$$

We prefer this formula due to lower memory required and lower numerical error.

Note 2: The circle inscribed within the unit square $[0, 1] \times [0, 1]$ is $(x-0.5)^2 + (y-0.5)^2 = 0.5^2$, not $x^2 + y^2 = 1$. Therefore, the judging condition whether the point belongs to circle should be

$$(x_i - 0.5)^2 + (y_i - 0.5)^2 \leq 0.5^2$$

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Problem Set 5

3. Every package of some intrinsically dull commodity includes a small and exciting plastic object. There are c different types of object, and each package is equally likely to contain any given type. You buy one package each day.
- (b) Find the mean number of days which elapse before you have a full set of objects.

3. Every package of some intrinsically dull commodity includes a small and exciting plastic object. There are c different types of object, and each package is equally likely to contain any given type. You buy one package each day.

(b) Find the mean number of days which elapse before you have a full set of objects.

Pay attention to the range of summation!

Let X_j denote the number of days which elapse between the acquisitions of the j th new type of object and the $(j + 1)$ th new type. From (a), we have $\mathbf{E}[X_j] = c/(c - j)$.

Let X denote the number of days which elapse before you have a full set of objects. We have

$$\mathbf{E}[X] = \mathbf{E} \left[\sum_{j=0}^{c-1} X_j \right] = \sum_{j=0}^{c-1} \frac{c}{c-j}$$

4. Each trial may result in any of t given outcomes, the i th outcome having probability p_i . Let N_i be the number of occurrences of the i th outcome in n independent trials.
- (b) Find the marginal mass functions of the multinomial distribution.

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(b) Find the marginal mass functions of the multinomial distribution.

No need to calculate summation from multinomial distribution!

Note that N_i just has the binomial distribution with parameters n and p_i .

$$\mathbf{P}(N_i = n_i) = \binom{n}{n_i} p_i^{n_i} (1 - p_i)^{n - n_i}$$

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4. Recitation



- Continuous random variables

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$

$$\mathbf{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$\text{var}(X) = \mathbf{E}[(X - \mathbf{E}[X])^2] = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$$

Density is not probability!

- Cumulative distribution functions

$$F_X(x) = \mathbf{P}(X \leq x) = \begin{cases} \sum_{k \leq x} p_X(k), & \text{discrete} \\ \int_{-\infty}^x f_X(t) dt, & \text{continuous} \end{cases}$$

- Properties of CDF F_X :

- F_X is monotonically nondecreasing:

$$\text{if } x \leq y, \text{ then } F_X(x) \leq F_X(y)$$

- $F_X(x)$ tends to 0 as $x \rightarrow -\infty$ and to 1 as $x \rightarrow \infty$.
- If X is discrete, then $F_X(x)$ is a piecewise constant function of x
- If X is continuous, then $F_X(x)$ is a continuous function of x
- If X is continuous, then

$$F_X(x) = \int_{-\infty}^x f_X(t)dt, \quad f_X(x) = \frac{dF_X}{dx}(x)$$

- Normal RV

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$$

- $\mathbf{E}[X] = \mu$, $\text{var}(X) = \sigma^2$
- $\frac{X - \mu}{\sigma} \sim N(0, 1)$
- LUT (Look Up Table) for CDF of standard normal

- Exponential RV

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- $\mathbf{E}[X] = \frac{1}{\lambda}$, $\text{var}(X) = \frac{1}{\lambda^2}$
- Memoryless property:

$$\mathbf{P}(X \geq c + x \mid X \geq c) = e^{-\lambda x} = \mathbf{P}(X \geq x)$$

1. Homework

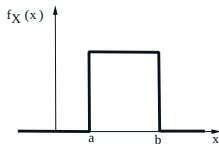
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3. Examples

4. Recitation



- **Uniform RV:** Consider a RV X that takes value in an interval $[a, b]$ with PDF



- Mean and variance:

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{a+b}{2}$$

$$\mathbf{E}[X^2] = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2 + ab + b^2}{3}$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{(b-a)^2}{12}$$

- **Exponential RV:** Consider an exponential RV X that has a PDF of the form

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- Mean and variance

$$\mathbf{E}[X] = \int_0^{\infty} x \lambda e^{(-\lambda x)} dx = (-x e^{-\lambda x}) \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = \frac{1}{\lambda}$$

$$\mathbf{E}[X^2] = \int_0^{\infty} x^2 \lambda e^{(-\lambda x)} dx = (-x^2 e^{-\lambda x}) \Big|_0^{\infty} + \int_0^{\infty} 2x e^{-\lambda x} dx = \frac{2}{\lambda^2}$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{1}{\lambda^2}$$

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1. Let Z be a continuous random variable with probability density function

$$f_Z(z) = \begin{cases} \gamma(1 + z^2), & \text{if } -2 < z < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) For what value of γ is this possible?
- (b) Find the cumulative distribution function of Z .

(a)

$$\int_{-2}^1 \gamma(1+z^2)dz = 1 \implies \gamma = \frac{1}{6}$$

(b)

$$F_Z(z) = \begin{cases} 1 & \text{if } z \geq 1 \\ \frac{1}{6}(z + z^3/3 + 14/3) & \text{if } 1 > z \geq -2 \\ 0 & \text{otherwise.} \end{cases}$$

2. The taxi stand and the bus stop near Al's home are in the same location. Al goes there at a given time and if a taxi is waiting, (this happens with probability $2/3$) he boards it. Otherwise he waits for a taxi or a bus to come, whichever comes first. The next taxi will arrive in a time that is uniformly distributed between 0 and 10 minutes, while the next bus will arrive in exactly 5 minutes. Find the CDF and the expected value of Al's waiting time.



2. The taxi stand and the bus stop near Al's home are in the same location. Al goes there at a given time and if a taxi is waiting, (this happens with probability $2/3$) he boards it. Otherwise he waits for a taxi or a bus to come, whichever comes first. The next taxi will arrive in a time that is uniformly distributed between 0 and 10 minutes, while the next bus will arrive in exactly 5 minutes. Find the CDF and the expected value of Al's waiting time.

Let A be the event that Al will find a taxi waiting or will be picked up by the bus after 5 minutes.

$$\mathbf{P}(A) = \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{2}$$

Y is the discrete part of X and has PMF

$$p_Y(y) = \begin{cases} 12/15 & \text{if } y = 0, \\ 3/15 & \text{if } y = 5. \end{cases}$$

Z is the continuous component of X has PDF

$$f_Z(z) = \begin{cases} 1/5 & \text{if } 0 \leq z \leq 5, \\ 0 & \text{otherwise.} \end{cases}$$

The CDF of X is

$$F_X(x) = \mathbf{P}(A)F_Y(y) + (1 - \mathbf{P}(A))F_Z(z)$$

and the expected waiting time is

$$\mathbf{E}[X] = \mathbf{P}(A)\mathbf{E}[Y] + (1 - \mathbf{P}(A))\mathbf{E}[Z] = \frac{15}{12}$$

3. Let λ be a positive number. The continuous random variable X is called exponential with parameter λ when its probability density function is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the cumulative distribution function (CDF) of X .
- (b) Find the mean of X .
- (c) Find the variance of X .
- (d) Suppose X_1 , X_2 , and X_3 are independent exponential random variables, each with parameter λ .
Find the PDF of $Z = \max\{X_1, X_2, X_3\}$.
- (e) Find the PDF of $W = \min\{X_1, X_2\}$.

(a)

$$F_X(x) = 1 - e^{-\lambda x}$$

(b)

$$\mathbf{E}[X] = \frac{1}{\lambda}$$

(c)

$$\text{var}[X] = \frac{1}{\lambda^2}$$

(d)

$$F_Z(z) = F_{X_1}(z)F_{X_2}(z)F_{X_3}(z)$$

$$f_Z(z) = \frac{dF_Z}{dz}(z)$$

(e)

$$F_W(w) = 1 - \mathbf{P}(W \geq w) = 1 - (1 - F_{X_1}(w))(1 - F_{X_2}(w))$$